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[5152]-178

S.E. (I.T.) (II Sem.) EXAMINATION, 2017

DATA STRUCTURES AND FILES

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer *four* questions.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. (a) Write an algorithm to convert an infix expression into postfix. [6]

(b) Write a Pseudo code to implement priority queue using multiple linked lists, one list for each priority for servicing patients in an hospital with priorities as : [6]

(i) Serious (top priority)

(ii) Medium illness (medium priority)

(iii) General (least priority).

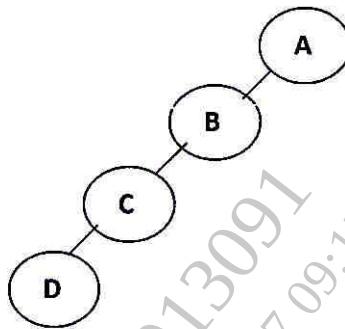
Or

2. (a) Write an algorithm to evaluate postfix expression. [6]

(b) Write a pseudo code to efficiently implement multiple queues in a single array. [6]

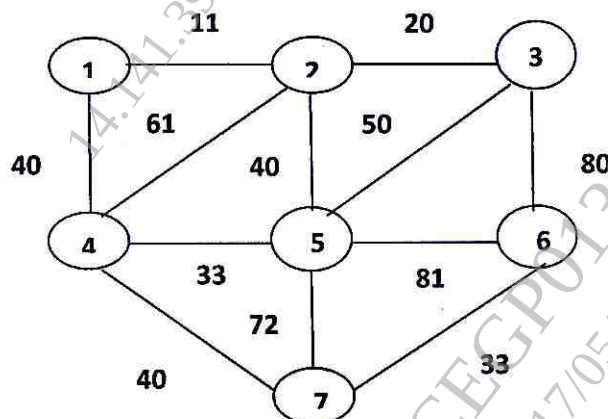
P.T.O.

3. (a) Write a non-recursive algorithm for inorder tree traversal. [4]
 (b) Consider the following specification of a graph G : [4]
 $V(G) = \{1, 2, 3, 4\}$
 $E(G) = \{(1, 2), (1, 3), (3, 3), (3, 4), (4, 1)\}$
 (i) Draw a picture of undirected graph.
 (ii) Draw its adjacency matrix.
 (c) Explain the array representation of binary tree using the following figure and explain the limitation of this representation. [4]



Or

4. (a) Write C++ function for insertion of a node into a BST. [4]
 (b) Construct a minimum spanning tree using Prim's algorithm for the given graph. [4]



- (c) Write a non-recursive algorithm for the DFS of a graph. [4]

5. (a) What is hashing ? Discuss about the characteristics of a good hashing function. [5]

(b) Sort the following number in ascending order using heap sort. Show all sorting steps : [5]

8, 20, 9, 4, 15, 10, 7, 22, 3, 12

(b) Suppose Max = 8 and Keys A, B, C, D have hash values Hash(A) = 3, Hash(B) = 0, Hash(C) = 4 and Hash(D) = 3. Use linear Probing for collision resolution. [4]

Or

6. (a) Draw a Huffman's tree for the given data set and find the corresponding Huffman's codes : [6]

Data	Frequency
P	18
Q	08
R	15
S	02
T	25
U	13
V	05
W	26

(b) Construct an AVL search tree by inserting the following elements in the order of their occurrence. Show the balance factor and type of rotation at each stage : [8]

MAR, MAY, NOV, AUG, APR, JAN,
DEC, JUL, FEB, JUN, OCT, SEP

7. (a) What is a File ? List various file opening modes. List the types of external storage devices. [6]
- (b) State the advantages, disadvantages and all primitive operations of sequential files. [6]

Or

8. (a) Compare sequential file organization with direct access file organization. [6]
- (b) Write a pseudo code to search a record from an index sequential file. [6]